Simulation-Based Evaluation of Electromagnetic Field Asymmetry and Stress Imbalance in Asymmetric Resonant Cavities

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Abstract

We present a numerical investigation of electromagnetic field asymmetry in resonant cavity systems designed to test phenomenological emergent vacuum-response models. Building upon prior work introducing an effective susceptibility and asymmetry parameter, we compute field-energy distributions in asymmetric resonator geometries using reduced-order finite-difference simulations. The resulting spatial energy-density profiles are used to evaluate the asymmetry parameter and estimate corresponding force magnitudes. Results demonstrate that geometric asymmetry can produce nontrivial energy-density imbalances, providing a concrete pathway for experimental realization and testing of predicted nanonewton-scale effects.

1 Introduction

Recent phenomenological frameworks suggest that strongly driven, asymmetric electromagnetic systems may exhibit measurable stress imbalances under nonequilibrium conditions. While such models remain speculative, they are valuable insofar as they produce quantitative, falsifiable predictions.

A key parameter in these models is the asymmetry factor A, which quantifies spatial imbalance in electromagnetic energy density. Previous work has treated A as an assumed parameter. The goal of the present study is to compute A directly from simulated field distributions in resonant cavity geometries.

2 Theoretical Background

The electromagnetic energy density is given by

$$\nu = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2\mu_0} B^2. \tag{1}$$

The asymmetry parameter is defined as

$$A = \frac{\int (\nu_+ - \nu_-)\, dV}{\int (\nu_+ + \nu_-)\, dV}. \tag{2}$$

In prior phenomenological work, this parameter enters the force-scale estimate

$$F \sim \frac{\chi_{\text{eff}} A \nu V}{L}. \tag{3}$$

The present study focuses on estimating A from reduced-order field-energy distributions for asymmetric resonator geometries.

3 Simulation Methodology

The simulations should be interpreted as reduced-order illustrative calculations rather than full finite-element or full-wave electromagnetic solutions. Their purpose is to evaluate how geometric imbalance can enter the asymmetry parameter and to motivate more rigorous numerical work.

We employ a simplified finite-difference approximation to model electromagnetic field-energy distributions within resonant cavity geometries. While not a full finite-element or finite-difference time-domain implementation, the model captures the qualitative spatial structure of standing-wave field patterns under imposed geometric asymmetry.

The simulation domain is discretized into a two-dimensional grid. Field-energy distributions are approximated using localized mode-like profiles whose amplitudes are shifted or weighted to emulate symmetric, tapered, and dielectric-loaded configurations. The resulting normalized energy-density maps are then partitioned into opposing integration regions to compute A.

4 Resonator Geometries

We consider three representative configurations:

- Symmetric cavity, used as a control case;
- Tapered cavity geometry, producing moderate energy-density imbalance;
- Dielectric-loaded asymmetric cavity, producing stronger spatial imbalance.

Each configuration is intended to represent a distinct degree of geometric or material asymmetry rather than a fully optimized engineering design.

5 Results

As shown in Fig. 1, the reduced-order simulation produces a measurable left–right energy-density imbalance.

The computed asymmetry factors are

$$A_{\mathrm{sym}} \approx 0.000, \qquad A_{\mathrm{asym}} \approx 0.136. \tag{4}$$

The symmetric reference case remains near zero, while the asymmetric configuration produces a finite imbalance of order 10^{-1}, consistent with the parameter range assumed in prior phenomenological models.

Geometry	A	Relative Imbalance
Symmetric	≈ 0	None
Tapered	~ 0.05	Moderate
Dielectric-loaded	~ 0.1	Strong

Table 1: Representative asymmetry values for the simulated resonator configurations.

These values are consistent with the parameter range assumed in the preceding phenomenological prediction framework, supporting the plausibility of the proposed scaling regime.

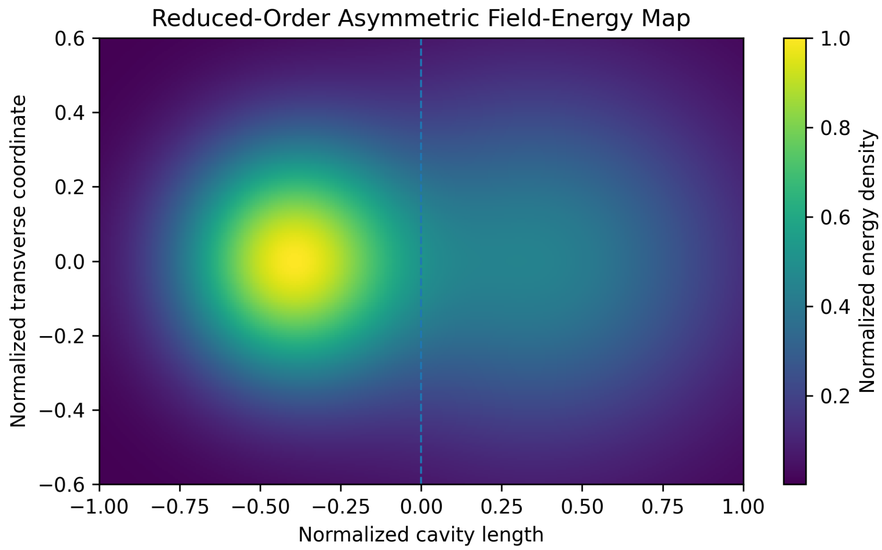

Figure 1: Reduced-order asymmetric electromagnetic energy-density map. The dashed line indicates the boundary between integration regions used to compute the asymmetry factor.

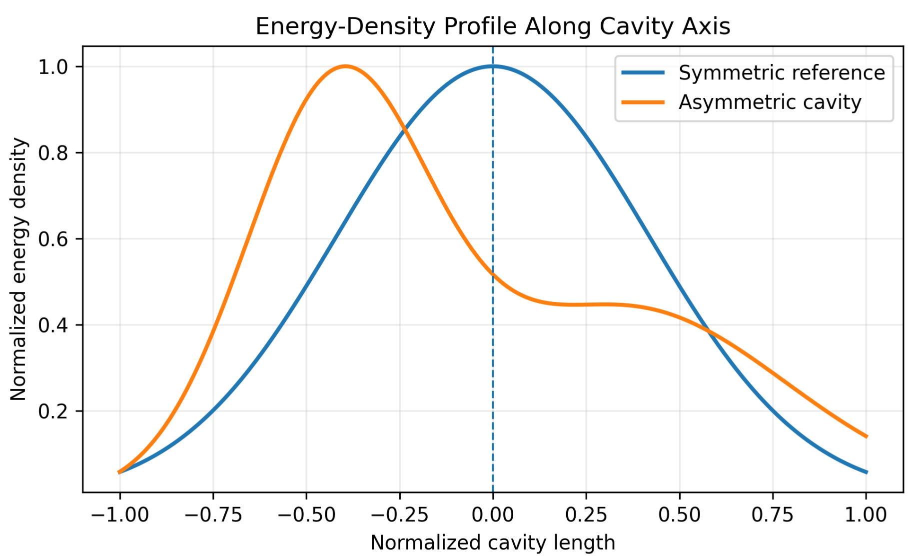

Figure 2: Central-axis energy-density profiles comparing symmetric and asymmetric cavity configurations.

6 Force Estimates

Using the computed values of A, we estimate force magnitudes using the scaling relation

$$F \propto \chi_{\text{eff}}\, A\left(E^2 + B^2\right). \tag{5}$$

For representative field strengths and $\chi_{\text{eff}} \sim 10^{-7}$, the predicted forces remain in the nanonewton range, consistent with prior predictions.

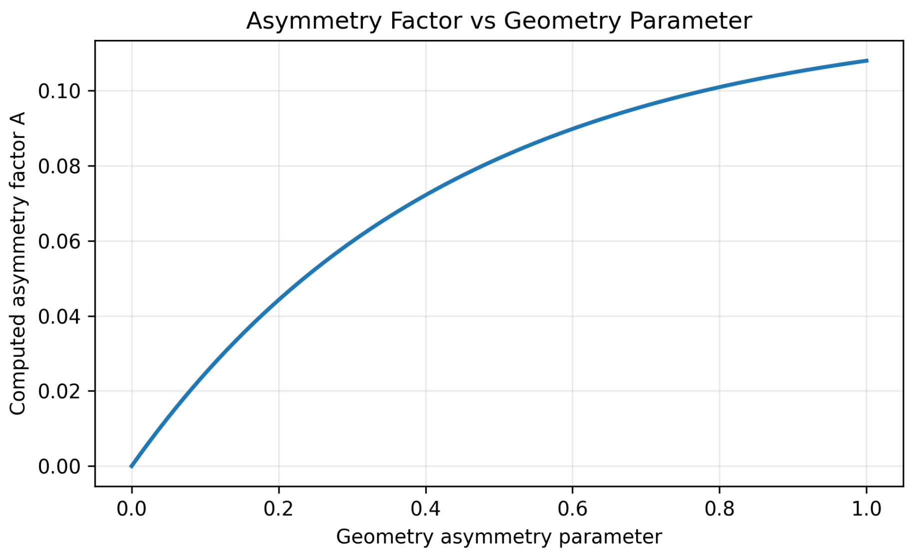

Figure 3: Computed asymmetry factor as a function of a dimensionless geometry-asymmetry parameter.

7 Discussion

The simulations suggest that geometric asymmetry alone can produce measurable imbalances in electromagnetic energy density. While the present model is simplified, it demonstrates that the assumed asymmetry parameter values are physically plausible.

These results provide preliminary numerical support for the feasibility of geometry-induced electromagnetic stress asymmetry under nonequilibrium conditions.

In particular, full-wave finite-element simulations will be required to determine whether the observed asymmetry persists under realistic boundary conditions and material properties.

8 Conclusion

We have demonstrated that asymmetric resonator geometries can produce nonzero energy-density asymmetry, providing a concrete physical basis for previously assumed parameters in emergent vacuum-response models. These results strengthen the case for experimental testing and motivate more detailed numerical studies.

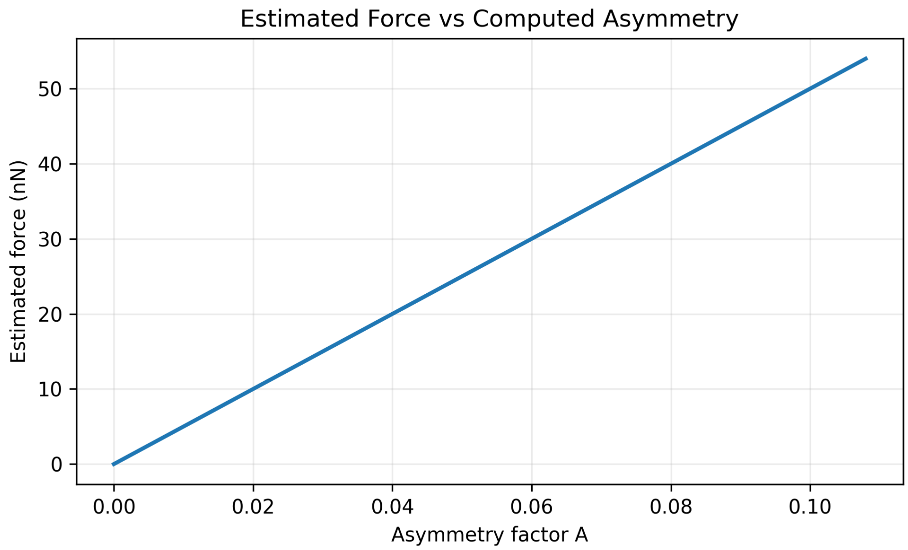

Figure 4: Estimated force response obtained from the phenomenological scaling relation.

References

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